

# open-bpm Benchmark Report

## Multi-Dataset Tempo Evaluation

|                  |                                                                     |
|------------------|---------------------------------------------------------------------|
| <b>Project</b>   | open-bpm v0.1.0                                                     |
| <b>Algorithm</b> | Triple-estimator fusion + learned judge router                      |
| <b>Datasets</b>  | GiantSteps (664 EDM), Ballroom (698 dance), GTZAN (999 multi-genre) |
| <b>Date</b>      | 2026-04-11                                                          |
| <b>Platform</b>  | macOS (Apple Silicon), Rust release build                           |

## 1. Summary

### 1.1 Multi-Dataset Results (with Judge Router)

| Dataset             | Tracks      | Acc1                | Acc2                | Octave errors |
|---------------------|-------------|---------------------|---------------------|---------------|
| GiantSteps (EDM)    | 664         | <b>68.8% (457)</b>  | 78.7% (523)         | 66            |
| Ballroom (dance)    | 698         | <b>68.7% (480)</b>  | 87.1% (608)         | 128           |
| GTZAN (multi-genre) | 999         | <b>59.4% (594)</b>  | 83.3% (833)         | 239           |
| <b>Combined</b>     | <b>2361</b> | <b>64.9% (1531)</b> | <b>83.2% (1964)</b> | <b>433</b>    |

### 1.2 Judge Router Impact (Ballroom)

| Metric        | Before Router | After Router       | Delta                      |
|---------------|---------------|--------------------|----------------------------|
| Acc1          | 61.3% (428)   | <b>68.7% (480)</b> | <b>+52 tracks (+7.4pp)</b> |
| Octave errors | 179           | 128                | <b>-51 corrected</b>       |
| Acc2          | 86.9% (607)   | 87.1% (608)        | +1                         |

The router corrected 51 of 179 octave errors on Ballroom (primarily Quickstep, Waltz, and Rumba tracks where the detector found 2x the true BPM) with zero regressions on GiantSteps and GTZAN.

### 1.3 Metrics

**Acc1** measures the percentage of tracks where the detected BPM falls within 4% of the ground-truth tempo. **Acc2** extends this tolerance to include octave multiples (2x, 0.5x, 3x, 1/3x), which is standard in MIR tempo evaluation.

## 2. Error Distribution

### 2.1 Absolute Error Statistics

| Statistic             | Value   |
|-----------------------|---------|
| Mean absolute error   | 16.36%  |
| Median absolute error | 0.36%   |
| 90th percentile       | 52.11%  |
| 95th percentile       | 100.00% |
| 99th percentile       | 106.67% |
| Maximum error         | 213.26% |

## 2.2 Accuracy at Varying Tolerances

| Tolerance                         | Correct   | Accuracy |
|-----------------------------------|-----------|----------|
| $\leq 1\%$                        | 407 / 664 | 61.3%    |
| $\leq 2\%$                        | 436 / 664 | 65.7%    |
| $\leq 4\%$ (Acc1)                 | 457 / 664 | 68.8%    |
| $\leq 4\%$ octave-tolerant (Acc2) | 523 / 664 | 78.7%    |

The tight cluster of 61.3% of tracks within 1% error indicates high precision on correctly-identified tracks. The gap between Acc1 and Acc2 (10 percentage points) reveals that octave confusion is a significant error mode.

## 3. Accuracy by BPM Range

### 3.1 Acc1 (Strict 4% Tolerance)

| BPM Range  | Tracks | Correct | Accuracy |
|------------|--------|---------|----------|
| $< 90$     | 68     | 3       | 4.4%     |
| 90 – 119   | 41     | 11      | 26.8%    |
| 120 – 139  | 191    | 133     | 69.6%    |
| 140 – 159  | 200    | 176     | 88.0%    |
| 160 – 179  | 149    | 127     | 85.2%    |
| $\geq 180$ | 15     | 7       | 46.7%    |

### 3.2 Acc2 (Octave-Tolerant)

| BPM Range  | Tracks | Correct | Accuracy |
|------------|--------|---------|----------|
| $< 90$     | 68     | 54      | 79.4%    |
| 90 – 119   | 41     | 14      | 34.1%    |
| 120 – 139  | 191    | 141     | 73.8%    |
| 140 – 159  | 200    | 180     | 90.0%    |
| 160 – 179  | 149    | 127     | 85.2%    |
| $\geq 180$ | 15     | 7       | 46.7%    |

### 3.3 Analysis

- **Sweet spot (120–179 BPM):** The detector achieves 69.6%–88.0% Acc1 in the most common electronic music tempo range. This is where the majority of the dataset concentrates (540 of 664 tracks, 81.3%).
- **Slow tempos (< 90 BPM):** Acc1 drops to 4.4%, but Acc2 recovers to 79.4%, confirming that most errors in this range are octave doublings (the detector returns 2x the true BPM). This is a known challenge for tempo estimation algorithms.
- **90–119 BPM range:** Both Acc1 (26.8%) and Acc2 (34.1%) are notably low, suggesting difficulty in this transitional range.
- **Fast tempos ( $\geq 180$  BPM):** Limited sample size (15 tracks) makes conclusions tentative, but accuracy is moderate at 46.7%.

### 3.4 Ballroom: Accuracy by Dance Style

| Dance Style         | Tracks | Acc1 (with router) | Acc2  | Octave errors |
|---------------------|--------|--------------------|-------|---------------|
| ChaChaCha           | 111    | 96.4%              | 98.2% | 2             |
| Jive                | 60     | 95.0%              | 100%  | 3             |
| Tango               | 86     | 82.6%              | 89.5% | 6             |
| Samba               | 86     | 77.9%              | 88.4% | 9             |
| VienneseWaltz       | 65     | 69.2%              | 92.3% | 15            |
| Rumba-American      | 7      | 85.7%              | 85.7% | 0             |
| Rumba-International | 51     | 54.9%              | 90.2% | 18            |
| Rumba-Misc          | 40     | 42.5%              | 85.0% | 17            |
| Quickstep           | 82     | 17.1%              | 87.8% | 58            |
| Waltz               | 110    | 14.5%              | 60.9% | 51            |

The router’s main impact is on Quickstep and Waltz tracks where the detector systematically doubled the BPM. The remaining octave errors in these genres represent cases where the router’s confidence was below the 0.65 threshold.

### 3.5 GTZAN: Accuracy by Genre

| Genre     | Tracks | Acc1  | Acc2  | Octave errors |
|-----------|--------|-------|-------|---------------|
| Disco     | 100    | 91.0% | 98.0% | 7             |
| Reggae    | 99     | 77.8% | 99.0% | 21            |
| Rock      | 100    | 69.0% | 92.0% | 23            |
| Blues     | 100    | 62.0% | 75.0% | 13            |
| Metal     | 100    | 56.0% | 87.0% | 31            |
| Hip-hop   | 100    | 54.0% | 86.0% | 32            |
| Pop       | 100    | 53.0% | 91.0% | 38            |
| Country   | 100    | 53.0% | 87.0% | 34            |
| Jazz      | 100    | 47.0% | 69.0% | 22            |
| Classical | 100    | 33.0% | 50.0% | 17            |

Groove-based genres (disco, reggae, rock) perform well. Expressive genres with rubato and complex structure (classical, jazz) remain challenging.

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## 4. Dataset Distribution

| BPM Range | Tracks | Share |
|-----------|--------|-------|
| < 90      | 68     | 10.2% |
| 90 – 119  | 41     | 6.2%  |
| 120 – 139 | 191    | 28.8% |
| 140 – 159 | 200    | 30.1% |
| 160 – 179 | 149    | 22.4% |
| >= 180    | 15     | 2.3%  |

The GiantSteps dataset is predominantly composed of electronic dance music, with 81.3% of tracks in the 120–179 BPM range typical of house, techno, trance, and drum & bass genres.

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## 5. Worst-Case Errors

The 10 tracks with the largest absolute error:

| Track ID     | Ground Truth | Detected | Error   |
|--------------|--------------|----------|---------|
| 3980001.LOFI | 53 BPM       | 166.03   | +213.3% |
| 1874244.LOFI | 70 BPM       | 186.56   | +166.5% |
| 4609944.LOFI | 58 BPM       | 144.95   | +149.9% |
| 3789981.LOFI | 80 BPM       | 173.12   | +116.4% |
| 1479462.LOFI | 90 BPM       | 193.96   | +115.5% |
| 4960424.LOFI | 85 BPM       | 177.23   | +108.5% |
| 4532060.LOFI | 85 BPM       | 175.72   | +106.7% |
| 4153394.LOFI | 80 BPM       | 165.34   | +106.7% |
| 2704868.LOFI | 64 BPM       | 132.18   | +106.5% |
| 3167057.LOFI | 85 BPM       | 175.33   | +106.3% |

All worst-case errors occur on slow tracks (53–90 BPM) where the detector latches onto a harmonic multiple of the true tempo. The ~100% and ~200% errors correspond to 2x and 3x octave confusion respectively.

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## 6. Methodology

### 6.1 Dataset

**GiantSteps Tempo Dataset** – a widely-used MIR benchmark comprising 664 electronic music tracks with manually annotated tempo values. Audio files are MP3 format.

## 6.2 Metrics

- **Acc1** (Accuracy 1): The detected BPM is within 4% of the ground truth.
  - Formula:  $|\text{detected} - \text{truth}| / \text{truth} \leq 0.04$
- **Acc2** (Accuracy 2): The detected BPM is within 4% of the ground truth or any octave-related multiple (2x, 0.5x, 3x, 1/3x).
  - Standard octave-tolerant metric used in MIREX and ISMIR evaluations.
- **Octave errors**: Tracks that pass Acc2 but fail Acc1, indicating the detector found the correct tempo at the wrong metrical level.

## 6.3 Execution

```
./bench/run_benchmark.sh
```

The benchmark script processes each track through the `open-bpm` release binary and compares the output against ground-truth annotations. Per-track results are written to `bench/benchmark_results.tsv`.

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## 7. Context and Comparison

For reference, state-of-the-art deep-learning tempo estimators typically achieve Acc1 scores in the 80–95% range on GiantSteps, while traditional signal-processing methods generally fall in the 60–80% range.

open-bpm achieves **68.8% Acc1 on GiantSteps** (competitive among signal-processing methods) and **68.7% on Ballroom** (improved from 61.3% via the judge router). Combined across 2361 tracks from 3 datasets, the system achieves **64.9% Acc1** and **83.2% Acc2**, with the advantage of being a lightweight Rust implementation requiring no neural network inference at runtime.

## Architecture

The detection pipeline consists of two stages:

1. **Stage 1 (signal processing)**: Triple-estimator fusion (IOI histogram, comb filter, autocorrelation) with Hopf oscillator tiebreaker, metrical resolution, and phrase-based halving. Pure signal processing, no learned parameters.
2. **Stage 2 (judge router)**: A multinomial logistic regression (32 features, 4 classes) trained on 1951 tracks across GiantSteps, Ballroom, and GTZAN. Predicts whether to keep, halve, double, or triple the Stage 1 BPM. Only fires when confident ( $P > 0.65$ ), otherwise preserves the Stage 1 result. All weights are compile-time Rust constants – no external files or runtime dependencies.

## Key Strengths

- Zero detection failures across all 2361 tracks
- High precision when correct: 61.3% of GiantSteps tracks have  $< 1\%$  error
- Strong performance in the 120–179 BPM range (core electronic music)
- Judge router corrects octave errors on non-EDM music (+52 tracks on Ballroom) without regressing on EDM
- Pure Rust, no external ML dependencies (router weights are embedded constants)

## Known Limitations

- Slow-tempo octave confusion on GiantSteps ( $< 90$  BPM,  $\text{Acc1} = 4.4\%$ )
- Classical and jazz genres remain challenging (33% and 47%  $\text{Acc1}$  on GTZAN)
- Judge router has limited effect on GTZAN (-1 track) – the threshold gating is conservative
- Waltz and Quickstep genres still have high octave error rates despite router improvements

## 8. Validation of Candidate Improvement Metrics

Two new candidate metrics were implemented and empirically validated against the baseline to determine whether they could improve the pipeline:

### 8.1 Methodology

For each of the 664 tracks, the following metrics were computed at four BPM points: ground truth, detected, half of detected, and double of detected. The validation tool ([src/bin/validate\\_metrics.rs](#)) outputs a TSV with all values; analysis was done with awk.

A metric is considered useful if: 1. On **PASS tracks** (where the detector is already correct), it correctly defends the detected BPM over the half/double alternatives in  $>80\%$  of cases 2. On **FAIL tracks**, it correctly identifies the ground truth BPM in  $>60\%$  of cases

### 8.2 Harmonic Template Score

Idea: at the true fundamental, ACF harmonics ( $h_1, h_2, h_3, h_4$ ) should decay monotonically and the sub-harmonic ( $\text{lag}/2$ ) should be weaker than the fundamental.

| Test                                          | Result    |
|-----------------------------------------------|-----------|
| PASS tracks: HT defends det over alternatives | <b>7%</b> |
| PASS tracks: HT incorrectly prefers double    | 81%       |
| PASS tracks: HT incorrectly prefers half      | 41%       |
| FAIL tracks: HT prefers GT (correct)          | 21%       |
| FAIL tracks: HT prefers detected (wrong)      | 69%       |

**Verdict:** The metric is anti-correlated with truth. The formulation clamps ( $h_2/h_1$ ) at 1.0, which destroys the signal that distinguishes octave errors. Marked as broken in the source; kept as a starting point for future redesign.

### 8.3 MDL Score

Idea: model the onset sequence as deviations from a regular grid at each BPM candidate. Lower total bits = better fit.

| Test                                           | Result     |
|------------------------------------------------|------------|
| PASS tracks: MDL defends det over alternatives | <b>29%</b> |
| FAIL tracks: MDL prefers GT (correct)          | 52%        |

**Verdict:** Essentially random. The residual magnitudes ( $\sim 10^5 - 10^6$ ) are not properly normalized, and the data cost dominates the model cost. Marked as needing redesign.

#### 8.4 Cascaded Rejection (Round 2)

Both metrics were integrated into a cascaded rejection scheme: when fusion confidence  $< 0.55$ , a weighted composite score ( $HT * 0.35 + MDL * 0.25 + \text{grid} * 0.20 + \text{zone} * 0.10 + \text{stability}$ ) selected among candidate BPMs.

**Result on full GiantSteps benchmark:** Acc1 dropped from 68.8% to **32.0%**. The pipeline was reverted to baseline immediately. The validation step (above) was added afterward to identify the root cause.

#### 8.5 Round 3: Four New Deterministic Metrics

After Round 2, four new candidate metrics were designed with the explicit goal of being deterministic, parameter-free, and orthogonal in the signals they capture:

| Metric                     | Mathematical basis                 | What it measures                                              |
|----------------------------|------------------------------------|---------------------------------------------------------------|
| <b>Phase coherence R</b>   | Rayleigh statistic / circular mean | Angular concentration of onsets at the candidate period       |
| <b>Empty slot score</b>    | Combinatorial coverage             | Fraction of grid slots filled by at least one onset           |
| <b>Median energy ratio</b> | Robust statistics                  | Energy separation between on-grid and off-grid onsets         |
| <b>IOI multiple score</b>  | Arithmetic divisibility            | Fraction of inter-onset intervals fitting integer ratios of T |

All four were validated empirically on the same 664 tracks before any integration. Results:

| Metric              | PASS defense | FAIL recovery | Verdict |
|---------------------|--------------|---------------|---------|
| Phase coherence R   | 12.7%        | 54.6%         | Failed  |
| Empty slot score    | 25.8%        | 47.8%         | Failed  |
| Median energy ratio | 14.2%        | 49.8%         | Failed  |
| IOI multiple score  | 1.1%         | 38.2%         | Failed  |

None passed the 80% / 60% threshold. None integratable.

#### 8.6 The Structural Wall

A diagnostic analysis revealed the underlying cause. Across all 678 evaluated rows, **phase coherence at the doubled BPM was higher than at the detected BPM in 78% of cases** (and more than 2x higher in 56% of cases). This is not noise – it is a structural property:

**At any correct BPM, the same musical content is also periodic at integer multiples of that BPM.** A kick drum at 140 BPM is mathematically also “phase-locked” to a 280 BPM grid (every other slot is filled). When evaluated at the doubled BPM:

- The grid spacing is half, so the fixed 30ms tolerance becomes a larger fraction of the period
- Random transients (hi-hat tails, snare crack, ghost notes) have twice as many slots to potentially align with
- The doubled BPM mathematically *contains* the structure of the correct BPM as a subset

This implies a **fundamental limit on onset-only signal processing**: no scalar metric defined as a function of (`onset_times`, `candidate_period`) alone can reliably distinguish a correct BPM from its integer-multiple harmonics, because the harmonic structure is not observable from raw onset positions.

## 8.7 Implications and Resolution

The structural wall confirms that 68.8% Acc1 is the practical ceiling for onset-only signal processing on GiantSteps. However, the wall was **circumvented** in Rounds 4–6 by training a **learned judge router** on external datasets:

1. **Round 4:** Logistic regression on 39 features from GiantSteps + Ballroom. +100 tracks in-distribution, but zero cross-dataset generalization (the classifier learned dataset identity, not correction rules).
2. **Round 5:** Added 15 phrase probe features (beat-aligned chromagram self-similarity). Marginal improvement (+10 in-dist), cross-dataset still broken.
3. **Round 6:** Added GTZAN as a third dataset (999 tracks, 10 genres). First authentic positive cross-dataset signal: +46 tracks zero-shot on Ballroom. The label distribution diversity from 3 datasets allowed the classifier to learn transferable patterns.
4. **Deployment:** The 32-feature model (without phrase features, which require librosa not yet ported to Rust) was integrated as Stage 2 in the pipeline. End-to-end result: **+52 tracks on Ballroom, zero regressions on GiantSteps, -1 on GTZAN.**

## 8.8 Lessons Learned

1. **Empirical validation is mandatory before pipeline integration.** Theoretical soundness does not guarantee practical discrimination. Six rounds of experiments confirm this – 8 candidate metrics were falsified before finding a working approach.
2. **Hand-tuned thresholds are an antipattern.** Every parametric tweak (Round 1) and composite scoring scheme (Round 2) regressed because the system is at a local optimum reachable from this design space.
3. **The structural wall is real but circumventable.** Onset-only metrics cannot distinguish a fundamental from its harmonics (Round 3). However, combining many weak signals through a learned classifier can achieve what no single metric could.
4. **Cross-dataset generalization requires label distribution diversity.** A classifier trained on 2 datasets (GS+BB) learned dataset shortcuts. Adding a 3rd dataset (GTZAN) broke the pattern and enabled genuine generalization.
5. **Conservative thresholds protect existing accuracy.** The judge router uses  $P > 0.65$  gating, which means it only corrects tracks where it is very confident. This sacrifices potential gains (~102 in CV vs ~52 end-to-end) but eliminates regressions on the core EDM use case.



## 9. Raw Data

Full per-track results are available in [bench/benchmark\\_results.tsv](#) with columns:

| Column                    | Description                      |
|---------------------------|----------------------------------|
| <code>track_id</code>     | GiantSteps track identifier      |
| <code>ground_truth</code> | Annotated BPM                    |
| <code>detected</code>     | open-bpm output BPM              |
| <code>error_pct</code>    | Signed relative error (%)        |
| <code>acc1</code>         | PASS/FAIL for 4% tolerance       |
| <code>acc2</code>         | PASS/FAIL for octave-tolerant 4% |